



Satish Hattekar <satishhattekar1997@gmail.com>

B28: SQL SuperHero Program : Session 22

1 message

Vishal Jaiswal <vishaldbdev@gmail.com>

21 March 2025 at 21:10

Bcc: satishhattekar1997@gmail.com

Question: can we write decode within decode or case within case. Like multiple checks ?

YES, we can see the example of decode in last notes, see here case

Case with in case

```
select first_name, last_name, DEPARTMENT_ID,  
       case when DEPARTMENT_ID = 90 then (case when JOB_ID = 'AD_PRES' then 'presiden' else 'normal job' end)  
       else 'normal job'  
       end as job_review  
from hr.employees
```

FIRST_NAME	LAST_NAME	DEPARTMENT_ID	JOB_REVIEW
Steven	King	90	presiden
Neena	Kochhar	90	normal job
Lex	De Haan	90	normal job
Alexander	Hunold	60	normal job
Bruce	Ernst	60	normal job
David	Austin	60	normal job
Valli	Pataballa	60	normal job
Diana	Lotney	60	normal job

```

select first_name, last_name, DEPARTMENT_ID, JOB_ID,
case when DEPARTMENT_ID = 90 and job_id = 'AD_PRES' then 'aaaa'
else 'normal job'
end as job_review
from hr.employees

```

↓ multiple when can use by using AND

FIRST_NAME	LAST_NAME	DEPARTMENT_ID	JOB_ID	JOB_REVIEW
Steven	King	90	AD_PRES	aaaa
Neena	Kochhar	90	AD_VP	normal job
Lex	De Haan	90	AD_VP	normal job
Alexander	Hunold	60	IT_PROG	normal job
Bruce	Ernst	60	IT_PROG	normal job
David	Austin	60	IT_PROG	normal job
Valli	Pataballa	60	IT_PROG	normal job
Diana	Lorentz	60	IT_PROG	normal job

=====

Question: can we write decode in group By ?

YES

```

SELECT DECODE(DEPARTMENT_ID,90,'Operations','60','Sales','all dept') Department,sum(salary) Salaries
from hr.employees
group by DECODE(DEPARTMENT_ID,90,'Operations','60','Sales','all dept');

```

```

select decode (department_id, 60, 'sales', 90, 'operations', 'all_dept') , sum(salary) from hr.employees
group by decode (department_id, 60, 'sales', 90, 'operations', 'all_dept')

```

↳ simple Aggregate fn concept.

DECODE(DEPARTMENT_ID,60,'SALES',90,'OPERATIONS','ALL_DEPT')	SUM(SALARY)
operations	58000
sales	28800
all_dept	605216

✓

3 rows selected.

```

select case when department_id = 90 then 'operations'
           when department_id = 60 then 'sales' else 'all_depat' end as dept_name , sum(salary) from hr.employees
group by case when department_id = 90 then 'operations'
           when department_id = 60 then 'sales' else 'all_depat' end

```

↳ same with case

DEPT_NAME	SUM(SALARY)
operations	58000
sales	28800
all_depat	605216

3 rows selected.

Concept is same. Group by column name and decode return a value, which can be used as column name

Can we Use case in Having Clause?

YES

```

select department_id, max(salary) maxsal
from hr.employees
group by department_id
having (max(case when department_id = 90 then salary end) > 20000
       or
       max(case when department_id <> 90 then salary end) > 10000
)

```

DEPARTMENT_ID	MAXSAL
110	12008
90	24000
30	11000
20	13000



What is having? Its a filter on aggregate. When we want to filter data on max, min, sum, count and avg. Suppose my situation is : if department_id is 90 then show me salary > 20000 and if department is <> 90 then > 10000

See the below, Update the employee table, if department is 90 the update the salary by 40 and if department is <> 90 then update by 20.
see , the condition comes , so case will work.

```
=====
```

```
update employees set salary = (case when DEPARTMENT_ID = 90 then salary+40
                                   when DEPARTMENT_ID <> 90 then salary+20
                                   end)
```

107 row(s) updated.

use in update

```
select * from employees
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY
100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24040
101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17040
102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17040
103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9020
104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6020
105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4820
106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4820

```
update emp set salary = decode(department_id, 90, salary+1000, 60, salary+2000)
```

107 row(s) updated.

when dept = 90 then sal f low
when dept = 60 then sal low

See below the use in where condition. I want to select data with filter condition that if department = 90 then show me those records where salary = 20000 and when 96 then 9000.
It means, filter records depends on department..

```
select employee_id, job_id, salary, department_id from employees
where salary > case when department_id = 90 then 20000
                    when department_id = 96 then 9000
                    else 10000
                    end
```

→ Simple use in
where clause

EMPLOYEE_ID	JOB_ID	SALARY	DEPARTMENT_ID
100	AD_PRES	24040	90
108	FI_MGR	12028	100
114	PU_MAN	11020	30
145	SA_MAN	14020	80
146	SA_MAN	13520	80
147	SA_MAN	12020	80
148	SA_MAN	11020	80
149	SA_MAN	10520	80